1.List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

1. Main components inside a desktop PC

No.	Component	Short description
1	Motherboard	The main circuit board that connects all computer components.
2	CPU (Processor)	Processes instructions and performs calculations.
3	RAM	Temporarily stores data used by running programs.
4	HDD	Stores files, software, and the operating system using magnetic disks.
5	SSD	Stores files and software using flash memory and provides faster access than an HDD.
6	Power Supply Unit (PSU)	Converts AC electricity into DC power for computer components.
7	CPU Cooler and Fan	Removes heat from the processor to prevent overheating.
8	Graphics Card (GPU)	Processes graphics and sends visual output to the display. It may be integrated into the CPU instead.
9	Cabinet/Case	Protects the internal components and holds them in place.
10	Case Fans	Improve airflow and help keep internal components cool.
11	Network Adapter	Connects the computer to a wired or wireless network. It may be built into the motherboard.
12	Sound Card	Handles audio input and output. It is usually integrated into the motherboard.
13	CMOS Battery	Maintains the real-time clock and certain BIOS/UEFI settings when power is disconnected.
14	Storage/Data Cables	SATA data cables connect compatible drives to the motherboard; power cables supply electricity.
15	Expansion Cards	Add features such as extra USB ports, networking, or other specialized functions.

**2.Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

Step	Action	Caption for your document
1	Unplug power	Turn off the PC and disconnect the power cable and all peripherals.
2	Remove side panel	Remove the cabinet screws and carefully take off the side panel.
3	Disconnect cables	Disconnect motherboard, storage, and other component cables carefully.
4	Remove RAM	Release the retaining clips and gently remove the RAM modules.
5	Remove HDD/SSD	Disconnect the drive cables and remove its mounting screws, if present.
6	Remove graphics card	Release the PCIe latch and screws, then carefully remove the card if installed.
7	Remove motherboard	Disconnect remaining cables, remove motherboard screws, and lift it carefully.
8	Inspect components	Place the components on a clean, non-conductive surface and inspect them.

3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step	Action	Explanation
1	Install motherboard	Align it with the case standoffs and secure it using the correct screws.
2	Install CPU and cooler	Install the processor and cooling system according to the motherboard manual, if they were removed.
3	Install RAM	Align the notch with the slot and press gently until the retaining clips lock.
4	Install HDD/SSD	Secure the drive and reconnect its data and power cables.
5	Install graphics card	Insert the card into the correct PCIe slot and secure it with screws.
6	Connect cables	Reconnect motherboard power, CPU power, storage, fans, and front-panel cables.
7	Close the cabinet	Check that all components and cables are secure, then replace the side panel.
8	Test the computer	Connect peripherals and power, start the PC, and verify that it boots successfully.

4.Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Tool	Why it is important
Phillips screwdriver	Removes most cabinet and component screws.
Flat-head screwdriver	Helps with compatible flat-head screws.
Anti-static wrist strap	Helps protect components from static electricity.
Small screw container	Keeps screws and small parts organized.
Flashlight	Helps identify connectors and screws inside the cabinet.
Cable ties	Organize cables and improve airflow.
Plastic spudger	Helps release suitable clips without scratching the case.
Clean, non-conductive work surface	Provides a safer area for handling components.